

Unit 6, Lesson 7: Triangle Similarity



Benchmark

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.



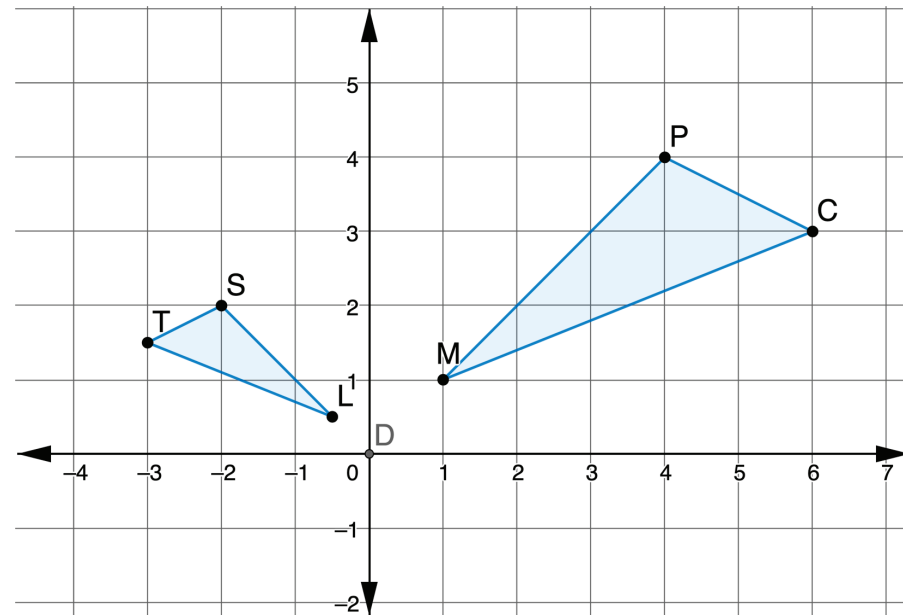
Learning Targets

- ☐ I can identify similar triangles and write similarity statements.
- ☐ I can use given information to show that two triangles are similar.



6.7.1 Warm-Up: Bell Work

- $\triangle MPC$ and $\triangle TSL$ are shown on the coordinate grid, where $\triangle MPC$ is the preimage.



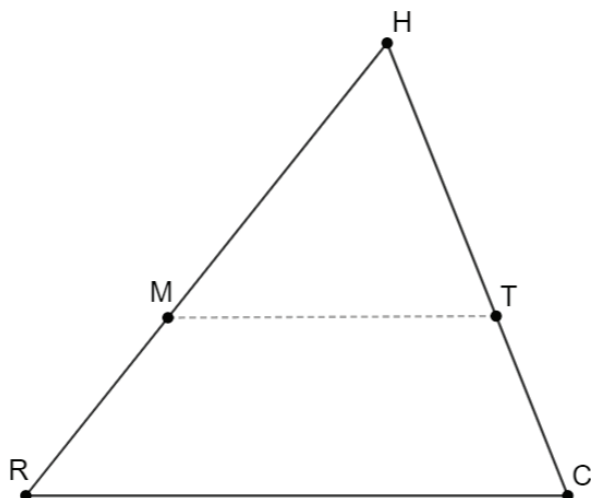
Describe a sequence of two transformations that will map the triangles.



6.7.2 Exploration: Similar Triangles

With your partner, complete the following.

$\triangle HCR$ is shown, where $m\angle HCR = 68^\circ$, $m\angle HRC = 51^\circ$, and $m\angle RHC = 61^\circ$. Point M is on $\overline{HR}$ and point T is on $\overline{HC}$. $\overline{MT}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.

Measurements of $\triangle RHC$		Measurements of $\triangle MHT$	
$RH =$	$m\angle HRC = 51^\circ$	$MH =$	$m\angle HMT =$
$HC =$	$m\angle RHC = 61^\circ$	$HT =$	$m\angle MHT = 61^\circ$
$RC =$	$m\angle HCR = 68^\circ$	$MT =$	$m\angle HTM =$

2. What do you notice about the corresponding angles of $\triangle RHC$ and $\triangle MHT$?

3. Discuss with your partner what relationship could have been given so that it would not have been necessary to use a protractor to find $m\angle HMT$ and $m\angle HTM$. Summarize your discussion.

4. What do you notice about the corresponding sides of $\triangle RHC$ and $\triangle MHT$?

5. Discuss with your partner how $\triangle RHC$ and $\triangle MHT$ are related. Summarize your findings.

6. Complete the statement that shows the relationship. $\triangle RHC$ is similar to $\triangle$ _____.

7. Complete each relationship for $\triangle RHC$ and $\triangle MHT$.

$$\frac{RH}{TM} = \frac{HT}{CR}$$

$$\frac{HT}{CR} = \frac{RC}{MT}$$

8. Discuss with your partner a sequence of transformations that can be used to map $\triangle RHC$ onto $\triangle MHT$. Summarize your discussion by describing the sequence.

9. Compare your sequence of transformations with another pair of students. Ask two classmates for their sequence of transformations. Record their sequence and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating your summary is correct.

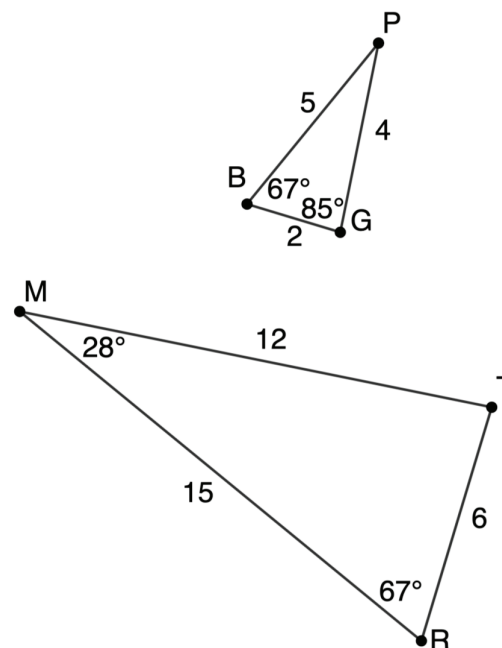
Sequence	My Summary of Their Reason	Initials

10. What transformation ensures that the sequence of transformations will result in similar triangles?



6.7.3 Your Turn!

1. $\triangle PGB$ and $\triangle MTR$ are shown. Use the given information to determine if $\triangle PGB$ is similar to $\triangle MTR$. If they are similar, write a similarity statement. If they are not, explain why.





6.7.4 Activity: Stations

1. With your partner, create two triangles using the description provided. Try to make the two triangles as different as possible.

Description Letter	Drawing

Description Letter	Drawing

Description Letter	Drawing

Description Letter	Drawing

2. Which descriptions resulted in similar triangles?



Angle-Angle Similarity (AA~) Theorem

If two angles of one triangle are congruent to two angles of another triangle, then the two triangles are similar.



Side-Side-Side Similarity (SSS~) Theorem

If the lengths of the corresponding sides of two triangles are proportional, then the triangles are similar.



Side-Angle-Side Similarity (SAS~) Theorem

If the lengths of two sides are proportional and their included angles are congruent on two different triangles, then the triangles are similar.

3. Match each description with the postulate or theorem that describes it. If none describe it, then write "none."

Description Letter	Theorem



6.7.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.